Sophia Bian

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EDUCATION

Northwestern University — Expected June 2029

B.S. Electrical Engineering, Minor in Computer Science — Evanston, IL

- Engineering Dean's Honors List (2025, 2026)
- Coursework: Computer Engineering Design · Intro to Computer Engineering · Object-Oriented Programming · Fundamentals of Computer Science · Linear Algebra · Multi-Variable Calculus

Henry M. Gunn High School — June 2025

High School Diploma — Palo Alto, CA

EXPERIENCE

Northwestern Formula Racing — September 2025 – Present

PCB & Firmware Engineer, DAQ Subteam — Evanston, IL

- Contributing to a ~25,000-line C++ firmware codebase on a dashboard team of 7 (subgroup of the DAQ subteam), building the driver-facing Dashboard Controller for a Formula SAE race car
- Designed V2 and V3 Dashboard Controller PCBs in Altium Designer: polygon pours for thermal management, high-current trace layout, and differential pair routing for CAN Bus signal integrity
- Implemented driver debugging screens and a Python script for automated screen asset generation, reducing manual firmware iteration overhead
- Maintained CAN Bus communication interfaces between the dashboard and vehicle subsystems across multiple firmware revisions

Querrey Simpson Institute for Bioelectronics — Rogers Research Group — January 2026 – Present

Undergraduate Research Assistant — Evanston, IL

- Assisting development of a portable wireless fluorimeter for continuous fluorescence monitoring in biomedical applications
- Assembled flexible PCB prototypes via SMD soldering; performed functional validation of photodetector sensing, power regulation, and Bluetooth communication subsystems
- Supported iterative hardware debugging and prototype refinement across multiple build cycles

PROJECTS

Resonance — Violin Posture Monitoring Wearable — 2025 – 2026

Northwestern Design Thinking & Communication

- Designed a wearable sensor unit using an Adafruit Feather RP2040 and LSM6DSOX + LIS3MDL 9-DoF IMU (accel/gyro/magnetometer) streaming at ~100 Hz over USB-C
- Modeled and 3D-printed a compact shoulder-mount housing (2.25 × 1.19 × 0.25 in, 18 g) for the PCB and rechargeable battery
- Built a no-install browser companion app using the WebSerial API with real-time 3D pose visualization, rolling-window posture scoring, and a post-session analytics dashboard

SPICALIGN — Assistive Medical Device — 2025 – 2026

Northwestern Design Thinking & Communication

- Designed a car-seat fitting aid for pediatric patients in hip spica casts, commissioned by Dr. Michelle Macy (Lurie Children's Hospital)
- Integrated foot-pedal-driven linear actuator (12–18 in travel) and continuous-rotation servo arms for hands-free operation during cast fitting

FIRST Robotics Competition — Team #1868 — 2022 – 2025

Electrical Subteam Lead — Palo Alto, CA

- Led wiring, PCB integration, and sensor systems for competition robots; FIRST World Championship qualifier ×2

SKILLS

Languages — C++, Python, MATLAB, HTML / CSS / JavaScript

Hardware	Altium Designer, NI Multisim, SMD soldering, PCB layout
Protocols	CAN Bus, Bluetooth, USB Serial / WebSerial, I²C, SPI
Mechanical	SOLIDWORKS, Fusion 360, OnShape
Tools	Git, GitHub

AWARDS & ACTIVITIES

- Engineering Dean’s Honors List — Northwestern University (2025, 2026)
- Aerospace & Aviation Sector Innovation Award — Conrad Challenge (2024)
- FIRST Robotics Team #1868 — Electrical Subteam Lead; World Championship qualifier ×2 (2022–2025)
- San Francisco Symphony Youth Orchestra — Flute & Piccolo (2023–2025); John Philip Sousa Award recipient